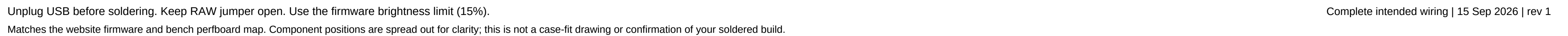
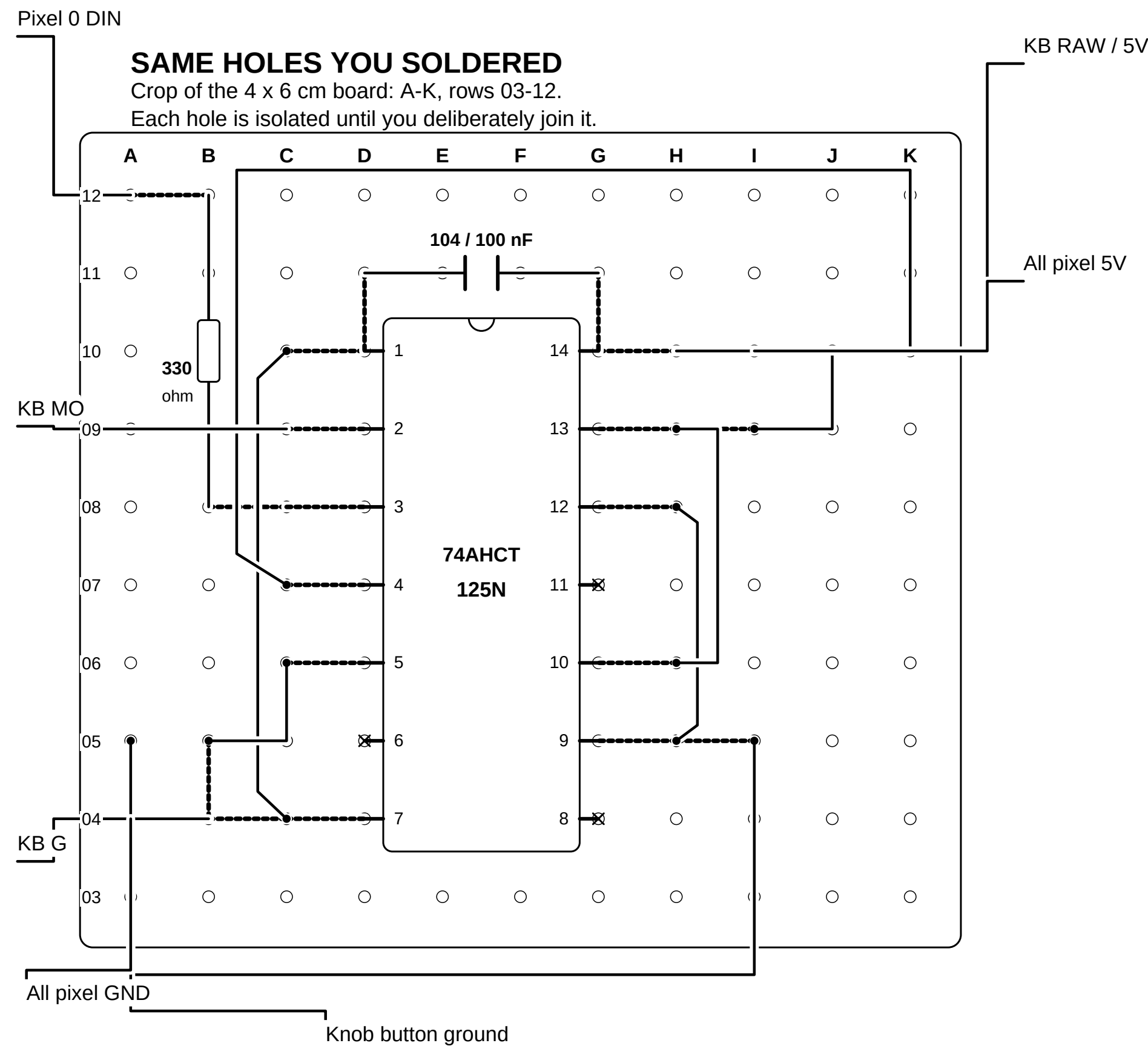


Expanded top-down view. Key positions match the front of the keyboard; hidden contacts are exposed as labeled insets.



Inside the perfboard

Top/component side. Column A is at the left; row 01 is at the bottom. Dashed bridges are on the underside, seen through the board.



NOTCH points toward the column letters / higher row numbers.

Looking underneath mirrors left and right. Locate the holes from above first.

Solid line = insulated wire on top. Short dashed line = bent lead + solder below.

Small X = intentionally unused chip output; leave unconnected.

HOW TO READ THE REST OF THE KEYBOARD

Keys

- A key closes a column-to-row path.
- Diode black band faces its row.
- Both Voice switches get a diode.

Knob + joystick

- Knob center of 3 terminals = GND.
- Other side: two pushbutton contacts.
- Joystick pads, from underside: X, 3V, Y, GND.

Lights

- All 5V pads share power; all GND pads share ground.
- Only DATA chains: OUT of one > IN of the next.
- Pixel pads: underside, arrows left to right, 5V at top.

References checked against this project and manufacturer documentation:

- KB2040: learn.adafruit.com/adafruit-kb2040/pinouts | Level shifter: ti.com/lit/ds/symlink/sn74ahct125.pdf
- NeoPixels: adafruit.com/product/4776 | Encoder: adafruit.com/product/377 | Joystick: sparkfun.com/thumb-slide-joystick.html
- Project source: `data/wiring.json` + `wiring-guide/connections.json` + `downloads/firmware/code.py` on codexmicro.diy.

WHAT EACH CHIP LEG DOES

SN74AHCT125N - the small chip translates only the LED data.

Pin	Hole	Function	Connect to
1	D10	/OE1	GND - enables channel 1
2	D9	1A input	KB2040 MO through C9
3	D8	1Y output	330 ohm resistor > A12 > pixel 0 DIN
4	D7	/OE2	5V - disables spare channel
5	D6	2A input	GND
6	D5	2Y output	Leave open
7	D4	GND	Common ground
8	G4	3Y output	Leave open
9	G5	3A input	GND
10	G6	/OE3	5V - disables spare channel
11	G7	4Y output	Leave open
12	G8	4A input	GND
13	G9	/OE4	5V - disables spare channel
14	G10	VCC	RAW / 5V through H10

THE TWO COMPONENTS ALREADY ON THE BOARD

- 330 ohm resistor: B8 to B12. B8-C8-D8 and A12-B12 are bridges.
- 104 ceramic capacitor: D11 to G11. D11-D10 and G11-G10 are bridges.
- The capacitor crosses power and ground; it does not short them together.

ADDITIONS FOR THE COMPLETE KEYBOARD

- 100 uF capacitor near the start of the light chain: + to 5V; stripe to GND.
- More lights: each needs 5V + GND, plus data from the preceding DOUT.
- Joystick, touch and indicators share GND. Use a separate soldered junction from the spare GND hole; the illustration does not assign new perfboard holes.